Revisiting Object-Centric Learning for Long-Tailed 3D Semantic Scene Graph Prediction

GiHyeon Kim, KunHo Heo, SuYeon Kim, and MyeongAh Cho*

***Abstract*—3D scene graph prediction has progressed rapidly with recent advances in deep learning. Prior work has largely framed its performance bottleneck as a predicate-centered long-tailed recognition problem and proposed numerous techniques accordingly. In this paper, we argue that this predicate-centered view can be misleading in closed-vocabulary benchmarks. Our earlier work revisited the role of object representations and showed that the discrimination power of 3D object features can substantially affect scene-graph performance. However, it did not explicitly account for the long-tailed nature of closed-vocabulary datasets. Through an empirical analysis, we show that the apparent long-tail behavior of predicates is strongly coupled with imbalance in object co-occurrence. This observation suggests that predicate imbalance can be mitigated from an object-centric perspective by explicitly modeling and correcting object-pair priors. Based on this insight, we propose a supervised contrastive learning scheme for robust 2D–3D feature alignment, and object-centric edge updating together with objects pair-wise re-balancing strategy to counteract co-occurrence bias during predicate estimation. Extensive experiments demonstrate that the proposed approach consistently improves Top-K mean Recall under long-tailed evaluation and yields measurable gains in 3D scene graph prediction performance.**

***Index Terms*—3D Scene Graph Prediction, 3D Scene Understanding, Representation Learning, 3D Point Cloud Processing, Long-Tail Distribution Problem.**

I. Introduction

3D SEMANTIC scene graph prediction aims to parse a 3D scene into a structured graph whose nodes represent object instances and whose directed edges represent semantic predicates between object pairs. This representation is a natural interface for downstream reasoning in VR/AR [1], robotics and embodied AI [2]–[5], as it bridges geometry, semantics, and relational context. Despite rapid progress, current closed-vocabulary benchmarks remain challenging due to severe long-tailed distributions: a small number of frequent predicates and object compositions dominate training, while rare relations and novel triplets are often poorly recognized.

A common view is to treat this issue primarily as a predicate-centered long-tail recognition problem, and accordingly devise debiasing objectives, re-balancing strategies, or stronger relation modules [6]–[12]. In parallel, substantial effort has been devoted to improving 3D scene graph prediction by strengthening object and relation representations through jointly optimized embedding space, self/weakly-supervision signal, or multi-modal cues [10], [13]–[15]. More recently, open-vocabulary extensions have leveraged vision–language models and large language models to enrich semantics and expand recognition beyond fixed label sets. While these directions have advanced the field, they often implicitly assume that better relation modeling (or better object descriptors) alone will be sufficient to resolve long-tailed predicate behavior.

In this work, extending our prior object-centric study [16], we revisit this assumption and show that it can be misleading in closed-vocabulary settings. Our empirical analysis reveals that even when object representations become substantially more discriminative, the mean-recall gains on predicates can remain marginal, indicating that predicate imbalance is not fully explained by object feature quality alone. Instead, we identify imbalance in object co-occurrence as a critical driver of long-tailed predicate behavior. Intuitively, predicate prediction is strongly conditioned on which object pair appears; when object posteriors are reliable, learning rare predicate signals effectively becomes object pair-conditioned, and optimization is increasingly dominated by head object pairs. This mechanism is illustrated in Fig. 1: improving tail-object features is necessary but insufficient unless the model also accounts for object-pair structure, which can determine whether a tail relation is correctly inferred.

Motivated by this perspective, we frame long-tailed predicate learning as an object-centric optimization problem that couples object representation quality, co-occurrence structure, and training bias. We then introduce simple yet effective mechanisms that explicitly incorporate object-pair priors into graph inference. First, we improve object discrimination under ambiguous categories via a triple-modal 2D-3D alignment scheme that fuses multi-view CLIP [17] visual cues into 3D object features through cross-attention, together with hard negative weighting to sharpen fine-grained separation. Second, we inject object-pair priors into relation inference by pair-conditioned edge updating, encouraging the GNN to learn pair-dependent relation representations. Third, we mitigate residual co-occurrence imbalance through object pair-wise re–balancing within the predicate objective, directly counteracting pair-frequency bias in the relationship triplet.

Our main contributions are summarized as follows:

- We provide empirical evidence that increasing object discrimination power alone does not necessarily resolve huge bias from head predicate under imbalanced 3D scene graph dataset, motivating a broader view beyond predicate-centric methods.
- We identify object co-occurrence imbalance as a central factor underlying predicate long-tail effects in closed-vocabulary 3D scene graph prediction, and provide a

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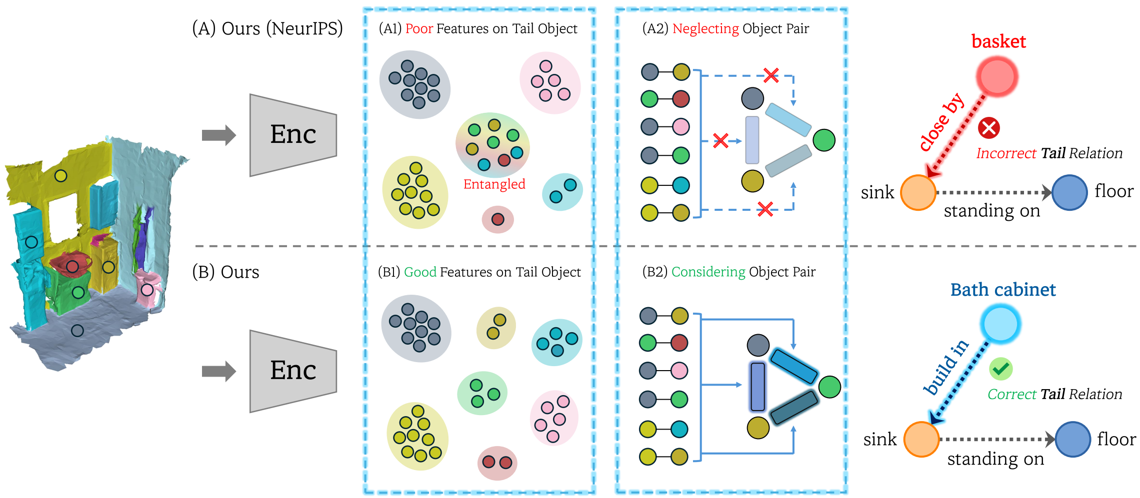

Fig. 1. **Problem Definitions**. Confusing hard-negative object categories and mischaracterized object-pair contexts can cause tail-predicate prediction errors, thereby substantially degrading performance under long-tailed distributions.

principled analysis linking reliable object posteriors to pair-conditioned optimization bias.

- We propose an object-centric solution that (i) strengthens object representations via triple-modal 2D–3D alignment with fine-grained negative weighting, and (ii) explicitly incorporates object-pair priors into predicate learning via pair-conditioned edge updating and object pair-wise Re–balancing.
- Extensive experiments demonstrate consistent improvements in Top-K mean Recall(mR@K), with particularly strong gains on tail predicates and unseen triplets, validating the effectiveness of the proposed object-centric regime.

II. Related Works

3D Point Cloud Representation. Early point-cloud representation learning focused on extracting permutation-invariant descriptors from raw points, exemplified by PointNet [18] and its hierarchical extension PointNet++ [19], which remain widely adopted as efficient backbones in 3D pipelines including 3DSSG. More recent progress has been driven by self-supervised and multimodal pretraining, where contrastive objectives improve transferability by aligning different views or modalities. For instance, PointContrast [20] learns robust features by matching local fragments across scans, and Hou *et al.* [21] incorporate scene-level context via graph-based contrastive learning. Cross-modal extensions further align point clouds with RGB images (CrossPoint [22]) and even unify language–image–point representations (CLIP^2 [23], ULIP/ULIP-2 [24], [25], MM-Point [26]), yielding strong generic 3D features under large-scale supervision. In contrast, our line of work [16] targets instance-level discriminability for robust closed-vocabulary 3D scene graph prediction, where improving the separability of semantically similar categories is critical for robust relational inference.

2D Scene Graph Prediction. In 2D scene graph generation and visual relationship detection [27]–[29], long-tailed and biased relation learning has been addressed through several complementary strategies. Several works [6], [7] model predicate dependencies and correlations to stabilize rare-predicate inference, designing debiased training objectives that explicitly mitigate frequency-driven bias beyond simple calibration. On the other hand, Tao et al. [30] strengthen the backbone with unified transformer-style formulations for richer relational reasoning. Recent work further expands relation semantics via multi-concept modeling and group-level relation detection, moving beyond strictly pairwise relations [8], [9]. Compared to 2D settings, 3D scene graph prediction must infer relations from geometry and 3D spatial context under imperfect perception, which introduces additional ambiguity beyond 2D appearance cues. While these approaches largely remain within a predicate-centric debiasing paradigm by reshaping predicate supervision or relation representations, Tao et al. [31] identify object label embeddings as a core bottleneck behind bias in 2D scene graph generation. Inspired by this insight, our work highlights a distinct and critical factor in an object-centric manner by revisiting the importance of object and its contextual signals on predicate estimation in heavily biased 3D scene graph.

3D Scene Graph Prediction. Early work such as SGPN [32] combined PointNet [18] with graph neural networks and established 3DSSG as a standard benchmark. Building on this, many methods refine node/edge representations by propagating information over adjacency graphs and injecting geometric or spatial priors [33]. Representative examples include attention-augmented GNNs and graph convolutions for relationship modeling [11], [34], as well as approaches that incorporate additional structural priors (e.g., via auto-encoding or auxiliary supervision) to enhance object and predicate features [10], [12]–[14]. Recent efforts further leverage RGB/RGB-D sequences and debiasing strategies to improve robustness under long-tailed distributions [35]–[37]. Complementary to these directions, our prior work [16] highlighted object representation quality as a key bottleneck; in this work, we show that stronger object features alone are insufficient to resolve long-tailed predicate behavior and identify object co-occurrence

TABLE I
CORE EMPIRICAL FINDING: IMPROVING OBJECT REPRESENTATION DISCRIMINABILITY DOES NOT NECESSARILY RESOLVE PREDICATE CLASS IMBALANCE OR SHOWS ONLY MARGINAL IMPROVEMENTS.

Experiment	Object			Predicate		PredCls	
	Acc@1	Balanced Acc	Macro-F1	mR@1	mR@3	mR@20	mR@50
OFL [16]	54.7	22.14	33.84	56.32	76.26	58.87	66.1
OFL [16] w/ PointNet++	56.5	24.99	34.94	56.85	77.23	59.61	66.05
ORD Enhancement	**60.1**	**25.11**	**39.31**	56.77	79.07	59.88	66.67

imbalance as a central driver of predicate imbalance under reliable object priors.

Open-Vocabulary 3D Scene Graph. Another active line of research extends 3D scene graph generation to open-vocabulary settings using VLMs/LLMs, leveraging their semantic priors for both object and relation modeling [38], [39]. Hybrid pipelines distill VLM-derived object and relation cues into 3D backbones and GNN predictors [40], while contrastive alignment objectives further bridge 3D representations with language supervision for open-vocabulary recognition [41]. In addition, LLM-centric systems construct graphs by detecting nodes from RGB-D observations and inferring edges via language-based reasoning, demonstrating strong transfer potential to downstream embodied tasks [2]. Subsequent studies continue to develop this paradigm by incorporating functional/interactive relations and broader downstream applicability [42]–[47].

We reiterate the significance of an object-centric regime, under which predicate class imbalance can be mitigated by explicitly accounting for object co-occurrence imbalance, particularly when the object representation space is sufficiently discriminative. This perspective elevates object co-occurrence from a secondary statistic to a first-class factor shaping predicate learning and long-tailed behavior. More broadly, our object-centric paradigm offers a unified lens for interpreting 3D scenes: it not only strengthens object understanding, but also provides a principled pathway to model relationships and contextual structure by grounding predicate inference in object-pair priors.

III. EMPIRICAL AND THEORETICAL FINDINGS

In our previous work [16], we didn't sufficiently concerned long-tailed distribution problem in 3DSSG dataset [32]. In this section, we compensate the empirical and theoretical observations in our previous work to find useful insights on 3D semantic scene graph prediction on severely long-tailed.

A. Empirical Observations

Predicate Long-Tail Distribution by Object Representation Discriminability. Table. I reports predicate inference performance in 3D scene graph prediction as a function of the class-wise discrimination power of 3D object features with GNN of our prior work [16]. The results indicate that improving Object Representation Discriminability alone is insufficient to fully address the long-tailed distribution in predicate classification. We compare two object feature encoders: (i) the NeurIPS baseline that increases ORD by adopting PointNet++, and (ii) an enhanced encoder that further boosts ORD through

TABLE II
NOTATION SUMMARY.

Symbol	Description
$o_i \in \mathcal{O}$	Object label (class) for node i
$r_{ij} \in \mathcal{R}$	Predicate (relation) label for edge (i, j)
$p_{ij} = \langle o_i, o_j \rangle$	Object pair (subject, object) associated with edge (i, j)
$\mathcal{P} = \mathcal{O} \times \mathcal{O}$	Set of all object pairs
z_i, z_j	Subject/object representations (object features) for nodes i and j
$P_{\text{data}}(r, p)$	Scene graph data distribution over predicate r and object pair p
$P_{\text{data}}(p)$	Marginal distribution of object pairs
$P_{\text{data}}(r \mid p)$	Conditional predicate distribution given an object pair

our additional contributions. Despite the clear improvement in object-level discrimination, the predicate metrics change only marginally across ORD levels, suggesting that higher ORD by itself is unlikely to resolve predicate imbalance.

To quantify Object Representation Discriminability (ORD), we crop per-object point clouds from the 3DSSG training and validation splits, extract object features, and train a linear classifier on top of the frozen features. We evaluate on the validation set using Top-1 Accuracy, Top-1 Balanced Accuracy, and Macro-F1. Based on these object-level metrics, we assign ORD levels and subsequently evaluate predicate performance using predicate classification mR@K as well as PredCls mR@K. Across ORD levels, predicate mean recall exhibits only marginal variations, revealing little correlation between object discrimination and long-tail robustness in predicate prediction. This observation suggests that the improvements attributed to the hypothesis in our prior NeurIPS work are better understood as gains in graph-level inference performance (e.g., SGCls and PredCls), rather than a direct resolution of class imbalance in predicate classification.

Predicate Distribution over Object Co-occurrence. Fig. 2 summarizes the statistics of object-pair co-occurrence and the associated predicate distribution. The blue bars show the number of distinct object pairs binned by their occurrence frequency in the training dataset, revealing a pronounced imbalance. Most object pairs appear only a few times (5 50 times), whereas a small subset of head pairs occurs very frequently (28 object pairs). This indicates that long-tailed problem in 3D scene graphs is not limited to object and predicate marginals, but also arises at the level of object-pair compositions within triplets.

In addition, the orange curve reports the median normalized

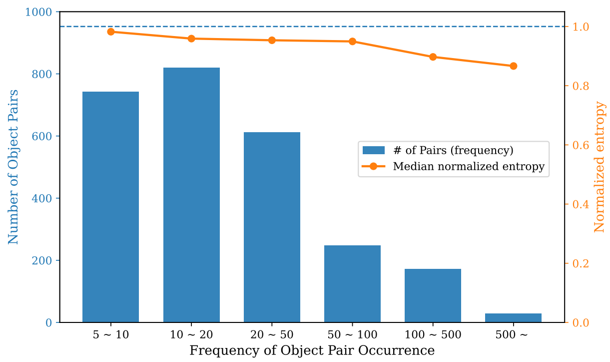

Fig. 2. **Empirical Findings of Object Pair Imbalance and Entropy of Predicate distribution**. Statistical Evidence of object pair imbalance and predicate distribution flatten out where they conditioned by object pairs – $P(r|o_i, o_j)$.

entropy of the conditional predicate distribution $P(r \mid o_i, o_j)$ for pairs in each frequency bin. Across all bins, the entropy remains close to 1, suggesting that once an object pair is given, the predicate distribution becomes relatively flat and diverse compared to the global predicate prior. In other words, object pairs act as a strong prior that reshapes predicate statistics, and the remaining ambiguity is distributed across multiple plausible relations rather than being concentrated on a single dominant predicate. These observations jointly motivate our object-centric view – methods that focus only on the marginal predicate imbalance $P_{\text{data}}(r)$ may overlook the dominant role of object pair frequency. Instead, explicitly accounting for object co-occurrence structure and incorporating object-pair priors into predicate inference provides a more direct handle for mitigating long-tailed predicate problem in 3D scene graph prediction. The detailed explanation about the settings of this observation study will be demonstrated in Appendix §.A.

B. Theoretical Findings

In this section, we provide a more principled analysis of how object co-occurrence influences biased predicate classification in 3D scene graph prediction, grounded in the empirical results presented above. In our prior work, we argued and empirically verified that predicate classification in 3D scene graphs implicitly or explicitly exploits object semantic labels as a strong prior. Specifically, given a ground-truth triplet label $\langle o_i, r_{ij}, o_j \rangle$, the predicate posterior can be expressed as

$$P_\theta(r_{ij}|z_i, z_j) = \sum_{o'_i, o'_j \in \mathcal{O}} P_\theta(r_{ij}|o'_i, o'_j) P_\theta(o'_i|z_i) P_\theta(o'_j|z_j) \quad (1)$$

where z_i, z_j, $\mathcal{O}$ are denoted in Table.2. This decomposition highlights that predicate prediction can be viewed as marginalizing over latent object labels inferred from the object features.

Building on this formulation, we extend our analysis and state two propositions. First, Proposition. (1), as Object Representation Discriminability (ORD) increases, the imbalance in object-pair frequencies becomes increasingly dominant in predicate classification, thereby reducing learning efficiency by biasing optimization toward head pairs. Second, Proposition. (2), when ORD is relatively low, the intrinsic class imbalance of predicates itself plays a more dominant role than object co-occurrence, since uncertainty in object identity weakens the effect of pair-conditioned priors.

To formalize these statements, we introduce three assumptions that characterize the data distribution and optimization dynamics. Evidence supporting these assumptions is provided in the Sec. III-A. Detailed notation is summarized in Table. II. We define the predicate learning objective as Eq.(2).

$$\mathcal{L}_{\text{pred}}(\theta) = \mathbb{E}_{(z_i, z_j, r)}[\ell(P_\theta(\cdot|z_i, z_j), r)] \quad (2)$$

We next introduce the formal definition of ORD, which will serve as a key quantity in the subsequent theoretical analysis.

Definition 1 (Object Representation Discriminability)**.** *Let $o \in \mathcal{O}$ denote the object label and z the corresponding object representation. We characterize Object Representation Discriminability (ORD) through the conditional entropy $H(o \mid z)$: higher ORD corresponds to lower uncertainty of o given z.*

To quantify the improvement in ORD, we define a non-negative indicator α as the reduction in conditional entropy with respect to a reference representation $z^{(0)}$:

$$\alpha \triangleq \frac{1}{H(o \mid z)}, \qquad \alpha \geq 0. \quad (3)$$

Thus, larger α indicates a larger decrease in $H(o \mid z)$ and, consequently, a stronger Object Representation Discriminability.

Assumption 1 (Long-tailed object-pair prior)**.** *The marginal distribution over object pairs, $P_{\text{data}}(p)$, is long-tailed; a small set of head pairs accounts for the majority of observations.*

Assumption 2 (Flattened conditional predicate distribution)**.** *For many object pairs p, the conditional predicate distribution $P_{\text{data}}(r \mid p)$ is less imbalanced than the global predicate distribution $P_{\text{data}}(r)$. Additionally, the tail predicate often happens with tail object pairs.*

Assumption 2 characterizes the relationship between the set of object pairs and the set of predicates. Intuitively, rare predicates tend to occur together with rare object labels. For example, `built in` may occur with the object pair ⟨ `sink`, `bath cabinet` ⟩, which is much rarer than a frequent triplet such as ⟨ `floor`, `standing on`, `table` ⟩. Based on this intuition, we assume that tail predicates mainly appear in certain tail object pairs, where the set of plausible predicate candidates is smaller than that of more frequent object pairs. Consequently, conditioning on a given object pair makes the predicate distribution relatively flatter, which in turn makes it feasible to leverage object-pair signals to better predict tail predicates.

Assumption 3 (Predicate-dominated optimization under high ORD)**.** *When Object Representation Discriminability is sufficiently high, the total objective is dominated by the predicate loss. Concretely, for*

$$\mathcal{L}_{\text{total}}(\theta) = \lambda_{\text{obj}} \mathcal{L}_{\text{obj}}(\theta) + \lambda_{\text{pred}} \mathcal{L}_{\text{pred}}(\theta), \quad (4)$$

we expect the optimization dynamics to satisfy

$$\left\|\nabla_\theta \mathcal{L}_{\text{pred}}\right\|_2 \gg \left\|\nabla_\theta \mathcal{L}_{\text{obj}}\right\|_2, \quad (5)$$

such that updates to θ are primarily driven by $\mathcal{L}_{\text{pred}}$ and the contribution of $\mathcal{L}_{\text{obj}}$ becomes marginal.

Under assumption. (3), we can neglect the impact of object classification loss while optimizing GNN network, focusing our interest on predicate loss function in theoretical analysis. The statistical evidence of these assumptions (1) and (2) is demonstrated in Sec. III-A.

Proposition 1 (Pair-frequency bias under pair-conditioned training)**.** *If the object posterior becomes one-hot as the representation power increases (α), i.e.,*

$$P_\theta(o \mid z) \xrightarrow[\alpha\to\infty]{} \delta_{o=\hat{o}(z)} \tag{6}$$

then the optimization in (2) *converges to a* pair-conditional *objective weighted by the pair frequency $P_{data}(p)$:*

$$\mathcal{L}_{\text{pred}}(\theta) \xrightarrow[\alpha\to\infty]{} \sum_{p\in\mathcal{P}} P_{data}(p)\,\mathbb{E}_{r\sim P_{data}(\cdot|p)}\big[\ell(P_\theta(\cdot \mid p), r)\big]. \tag{7}$$

Consequently, the mini-batch gradient is dominated by head pairs, and the mismatch between the frequency-weighted objective (7) *and a balanced objective aligned with mean recall yields an increased* pair-frequency gradient bias.

Proof. As α gets larger, we have $P_\theta(o|z_i) \to \delta_{o=o_i}$. Substituting this into (1) gives

$$P_\theta(r|z_i, z_j) \xrightarrow[\alpha\to\infty]{} P_\theta(r \mid o_i, o_j). \tag{8}$$

Therefore, (2) converges to

$$\begin{aligned}\mathcal{L}_{\text{pred}}(\theta) &= \mathbb{E}_{(p,r)\sim P_{\text{data}}}\big[\ell(P_\theta(\cdot \mid p), r)\big] \\ &= \sum_{p\in\mathcal{P}} P_{\text{data}}(p)\,\mathbb{E}_{r\sim P_{\text{data}}(\cdot|p)}\big[\ell(P_\theta(\cdot \mid p), r)\big],\end{aligned} \tag{9}$$

which is exactly (7). Since the coefficient is $P_{\text{data}}(p)$, head pairs (with large $P_{\text{data}}(p)$) dominate the objective and thus dominate the mini-batch gradients under assumption (2).

Next, consider an perfectly balanced objective which can completely resolves long-tail problem

$$\mathcal{L}_{\text{bal}}(\theta) = \sum_{p\in\mathcal{P}} w(p)\,\mathbb{E}_{r\sim P_{\text{data}}(\cdot|p)}\big[\ell(P_\theta(\cdot \mid p), r)\big], \tag{10}$$

where $w(p)$ is a balancing weight (e.g., $w(p) \propto 1/P_{\text{data}}(p)$). By linearity of differentiation and expectation, the gradient mismatch between (9) and (10) is

$$\begin{aligned}&\nabla\mathcal{L}(\theta) - \nabla\mathcal{L}_{\text{bal}}(\theta) = \\ &\quad\sum_{p\in\mathcal{P}} \big(P_{\text{data}}(p) - w(p)\big)\,\nabla\mathbb{E}_{r\sim P_{\text{data}}(\cdot|p)}\big[\ell(P_\theta(\cdot \mid p), r)\big].\end{aligned} \tag{11}$$

As $\alpha \to \infty$, the collapse (8) becomes exact, hence the form (11) precisely characterizes the induced *pair-frequency gradient bias* which leads erroneous training process with tailed object pair. □

Proposition 2 (Low representation power induces predicate-frequency gradient bias)**.** *Assume that the object posterior becomes sufficiently diffuse as the discrimination power decreases (a.k.a, α is large), so that for most inputs the*

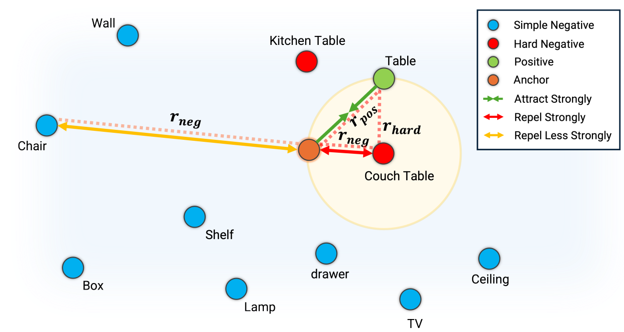

Fig. 3. **Negative sample selection and weighting in the embedding space.** Negatives highly similar to the positive ($r_{\text{hard}} > \beta$) are selected as hard negatives and repelled more strongly from the anchor via an exponential weight, encouraging finer separation among semantically close categories.

pair posterior $P_\theta(p \mid z_i, z_j)$ is close to an input-independent prior $\bar{P}(p)$. Then $P_\theta(r \mid z_i, z_j)$ degenerates toward an input-agnostic global predictor

$$P_\theta(r \mid z_i, z_j) \approx \sum_{p\in\mathcal{P}} P_\theta(r \mid p)\,P_\theta(p) \triangleq \bar{q}_\theta(r), \tag{12}$$

and the optimization in (2) *is approximately dominated by the* global *predicate frequency $P_{data}(r)$:*

$$\mathcal{L}_{\text{pred}}(\theta) \approx \sum_{r\in\mathcal{R}} P_{data}(r)\,\ell(\bar{q}_\theta(\cdot), r). \tag{13}$$

Consequently, the induced gradient follows a predicate-frequency weighting, *yielding a* predicate-frequency gradient bias *relative to a ideally balanced objective, which directly harms tail-predicate Top-K mean recall.*

Proof. Under the diffuseness assumption, $P_\theta(o_i, o_j \mid z_i, z_j) \approx \bar{P}(p)$ for most inputs, hence substituting into (1) yields (12). Plugging (12) into (2) gives

$$\begin{aligned}\mathcal{L}_{\text{pred}}(\theta) &= \mathbb{E}_{(z_i,z_j,r)\sim P_{\text{data}}}\big[\ell(P_\theta(\cdot \mid z_i, z_j), r)\big] \\ &\approx \mathbb{E}_{r\sim P_{\text{data}}}\big[\ell(\bar{q}_\theta(\cdot), r)\big] = \sum_{r\in\mathcal{R}} P_{\text{data}}(r)\,\ell(\bar{q}_\theta(\cdot), r),\end{aligned} \tag{14}$$

which is (13).

By linearity of differentiation and expectation, the corresponding gradient satisfies

$$\nabla\mathcal{L}(\theta) \approx \sum_{r\in\mathcal{R}} P_{\text{data}}(r)\,\nabla\ell(\bar{q}_\theta(\cdot), r). \tag{15}$$

Now consider an ideally balanced objective

$$\mathcal{L}_{\text{mR}}(\theta) = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} \ell(\bar{q}_\theta(\cdot), r). \tag{16}$$

Then the gradient mismatch between (15) and (16) is

$$\nabla\mathcal{L}(\theta) - \nabla\mathcal{L}_{\text{mR}}(\theta) \approx \sum_{r\in\mathcal{R}} \left(P_{\text{data}}(r) - \frac{1}{|\mathcal{R}|}\right)\nabla\ell(\bar{q}_\theta(\cdot), r), \tag{17}$$

which explicitly shows that the optimization is biased toward frequent predicates. When $P_{\text{data}}(r)$ is long-tailed, tail predicates contribute little to the update, leading to degraded tail-predicate recall and thus lower mean recall. □

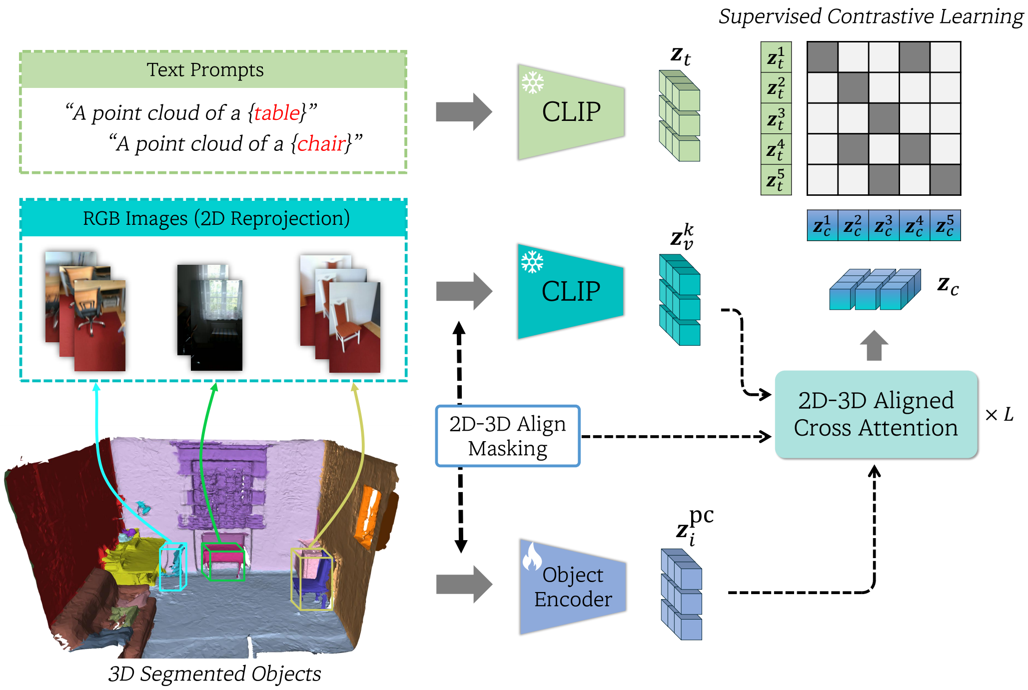

Fig. 4. **Overview of the object encoder pretraining architecture**. The method fuses three modalities—3D point clouds, 2D RGB images via CLIP visual features, and text prompts via CLIP text embeddings—through a cross-attention module with 2D-3D alignment masking. The resulting triple-modal representations are trained using supervised contrastive learning to enhance Object Representation Discriminability.

In summary, the two propositions above lead to the following conclusions:

1) When Object Representation Discriminability is sufficiently high, resolving predicate imbalance requires addressing object co-occurrence imbalance as well; improving object representations alone is not enough to yield balanced predicate inference.
2) To overcome object co-occurrence imbalance and effectively mitigate predicate long-tail problem, two components are essential: (i) alleviating object class imbalance, and (ii) explicitly incorporating object co-occurrence as a prior in the predicate inference process.

To reflect these conclusions, we propose three methods: (i) *Object Representation Discriminability Enhancement*, (ii) *Object-Centric Edge Updating*, and (iii) *Object Pair Inverse Frequency Weighting*.

IV. Proposed Method

Problem Formulation. Given an indoor 3D scene, our objective is to predict a semantic scene graph represented as a directed graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V}$ and $\mathcal{E}$ denote the sets of object nodes and relational edges, respectively. The input is a 3D point cloud $\mathbf{P} \in \mathbb{R}^{N\times 3}$ with N points, together with a set of class-agnostic instance masks $\mathcal{M} = \{M_k\}_{k=1}^{K}$ that partition the point cloud into K object instances. Following the 3DSSG protocol, the point cloud is reconstructed from 3RScan [48], which provides RGB-D image sequences during scene capture. Each instance node $\hat{o}_i \in \mathcal{V}$ is associated with a ground-truth semantic label $o_i \in \mathcal{O}$, where $\mathcal{O}$ is the predefined set of object categories. For any ordered pair of instances $(\hat{o}_i, \hat{o}_j)$, an edge $e_{ij} \in \mathcal{E}$ encodes the semantic predicates between them, forming a triplet $\langle \texttt{subject}, \texttt{predicate}, \texttt{object} \rangle$ in which $\hat{o}_i$ acts as the subject and $\hat{o}_j$ as the object. Predicate annotations follow a multi-label setting: each edge may be associated with one or more relation labels $r_{ij} \in \mathcal{R}$ drawn from the predicate vocabulary $\mathcal{R}$ (e.g., `standing on`, `hanging in`).

A. Object Representation Discriminability Enhancement

In our previous work [16], we introduced an object feature learning (OFL) method to enhance the discrimination power of 3D object representations. However, the approach relied on supervised contrastive learning with CLIP [17] visual features, where 2D–3D and text-3D alignment was performed solely based on class labels. This label-only supervision leads to two key limitations. First, even among 3D objects sharing the same label, substantial intra-class variations (e.g., shape, size, and geometric details) exist, and forcing them to be aligned can act as noise rather than useful supervision. Second, semantically similar categories may be assigned different labels depending on subtle functional or design cues. For instance, the distinction between chair and armchair may be marginal in the CLIP [17] embedding space compared to categories with clearly different semantics (e.g., table), making strict label-based alignment less effective.

To address these issues and further improve Object Representation Discriminability (ORD), we propose a triple-modal alignment strategy that combines a 2D–3D alignment module based on cross-attention with a modified hard-negative weighting scheme. The core idea is to avoid treating all same-label pairs as equally positive, and instead learn a more instance-

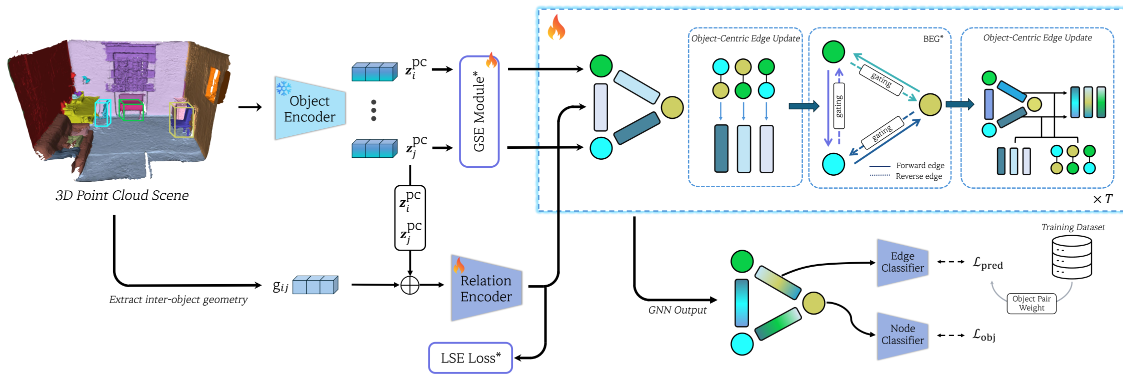

Fig. 5. **Overall architecture of the proposed 3D scene graph prediction method.**[1] The pipeline consists of an object encoder that extracts 3D point cloud features, a relation encoder that computes inter-object geometric features, and a GNN module with object-centric edge updating (OEU) via gating and pair-conditioned modulation. The predicate classification loss $\mathcal{L}_{\text{pred}}$ incorporates object pair inverse frequency weighting (OIW) to mitigate co-occurrence bias.

aware alignment that exploits multi-view visual observations while reducing supervision noise.

Triple Modal Cross Attention. As illustrated in Fig. 5, unlike previous work that directly uses CLIP [17] visual features for contrastive supervision, we incorporate them through a cross-attention module to produce visually informed 3D object representations. Since the 3RScan [48] dataset provides point clouds reconstructed from RGB-D sequences, we can associate each 3D object point cloud with the set of RGB-D frames that observe it as mentioned above. Let $z_i^{\text{pc}} \in \mathbb{R}^D$ denote the feature of the i-th 3D object from object encoder, and let $\{z_{i,k}^{\text{vis}}\}_{k=1}^{K_i}$, $z_{i,k}^{\text{vis}} \in \mathbb{R}^D$, be the CLIP [17] visual features extracted from the K_i matched RGB images that view object $\hat{o}_i$. We perform cross-attention from 3D object features as queries to their corresponding visual features as keys/values. To support batching with a variable number of views per object, we introduce an attention mask $M \in \mathbb{R}^{B\times T}$, where B is the batch size and $T = \sum_{i=1}^{B} K_i$ is the total number of visual tokens in the batch. After passing through L stacked cross-attention layers, the resulting fused representations are used for supervised contrastive learning against the CLIP [17] text features z_i^{text}. The cross attention operation is defined as

$$\alpha = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V^\top \odot M, \tag{18}$$

where $Q = \left[z_1^{\text{pc}}, z_2^{\text{pc}}, \ldots, z_B^{\text{pc}}\right] \in \mathbb{R}^{B\times D}$ and the keys and values are constructed from the concatenation of all visual features in the batch: $K = V = \left[z_{1,1}^{\text{vis}}, z_{1,2}^{\text{vis}}, \ldots, z_{B,K_B}^{\text{vis}}\right] \in \mathbb{R}^{T\times D}$, $T = \sum_{i=1}^{B} K_i$. To prevent visual features of one object from attending to another object's query, we apply an 2D-3D align mask. Specifically, the mask is defined to allow query (object features) to attend only to the subset of visual tokens corresponding to object, i.e., $M = \mathbb{1}\left[\sum_{l=1}^{K_{i-1}} K_l \leq j \leq \sum_{l=1}^{K_i} K_l\right]_{1\leq i\leq K, 1\leq j\leq T}$, which blocks all cross-object interactions within a batch.

This design enables instance-aware fusion of multi-view 2D cues into 3D object representations, mitigating the noise induced by naive label-based alignment and providing a stronger foundation for improving ORD under long-tailed object distributions.

Fine-Grained Negative Reweighting. As illustrated in Fig. 3, the object taxonomy of 3DSSG contains many semantically close categories such as `couch table` and `table`, whose CLIP [17] embeddings are substantially harder to separate than those of clearly distinct classes (e.g., `TV`). Such ambiguity in the CLIP [17] feature space can weaken label-supervised alignment and limit the discriminative power of learned 3D object representations, which is critical for tail object classification. To explicitly encourage separation among these confusing categories and leverage theoretical findings in Sec. III-B, we introduce a simple yet effective Fine-Grained Negative Reweighting(FGNR) strategy that improves Object Representation Discriminability (ORD) under a supervised contrastive learning setting.

Given a negative and positive sample $z_{\text{neg}}, z_{\text{pos}}$, we compute their similarity like $r_{\text{hard}} = \mathrm{sim}(z_{\text{pos}}, z_{\text{neg}})$. If the similarity exceeds a fixed threshold $\beta = 0.7$, we treat the negative as a hard negative and assign it an exponentially increasing weight. Specifically, the weight is defined as

$$w_i = \begin{cases} \exp(r_{\text{neg}} - r_{\text{hard}}), & \text{if } r_{\text{hard}} > \beta \\ 0, & \text{otherwise.} \end{cases} \tag{19}$$

where $r_{\text{neg}} = \mathrm{sim}(z_i^{\text{pc}}, z_{\text{neg}})$ denotes the cosine similarity distance between anchor and hard negative sample. This design assigns larger weights to negatives that are closer to the anchor than positives in the embedding space, thereby prioritizing the most confusing negatives during optimization.

We incorporate the proposed weights into the supervised contrastive objective by reweighting the negative terms in the denominator. For each anchor i with a positive sample z_{pos} and a set of negatives $\mathcal{N}(i)$, the weighted contrastive loss is computed as

$$\mathcal{L}_i^{\text{cross}} = -\log \frac{\exp \mathrm{sim}(z_i^{\text{pc}}, z_p)/\ \tau}{\sum_{r\in\mathcal{N}(i)} w_r \exp \mathrm{sim}(z_i^{\text{pc}}, z_r)/\ \tau} \tag{20}$$

where $\tau = 0.07$ is the temperature parameter. By emphasizing hard negatives while suppressing easy ones, this

[1]* that this module is the same as in our prior work [16]

TABLE III
QUANTITATIVE RESULTS (%) OF THE SGCLS AND PREDCLS TASKS WITH GRAPH CONSTRAINTS ON 3DSSG DATASET WITH 160 ENTITY AND 27 RELATION CLASSES.

Method	SGCls R@K			PredCls R@K			SGCls mR@K			PredCls mR@K		
	R@20	R@50	R@100	R@20	R@50	R@100	mR@20	mR@50	mR@100	mR@20	mR@50	mR@100
SGPN [32]	27.0	28.8	29.0	51.9	58.0	58.5	19.7	22.6	23.1	32.1	38.4	38.9
Zhang et al. [12]	28.5	30.0	30.1	59.3	65.0	65.3	24.4	28.6	28.8	56.6	63.5	63.8
SGFN [34]	29.5	31.2	31.2	65.9	78.8	79.6	20.5	23.1	23.1	46.1	54.8	55.1
VL-SAT [13]	32.0	33.5	33.7	67.8	79.9	80.8	31.0	32.6	32.7	57.8	64.2	64.3
HSLC [49]	–	32.2	32.5	–	68.2	68.3	–	29.6	29.8	–	63.9	64.0
CCL-3DSG [41]	**37.6**	40.3	**45.7**	**73.6**	79.9	82.9	35.0	37.3	**40.6**	59.1	66.7	69.1
Feng et al. [36]	–	**41.5**	42.3	–	**85.5**	85.9	–	**39.1**	39.8	–	**70.3**	**70.6**
Feng et al. [35]	–	37.1	38.7	–	85.4	**86.1**	–	35.3	36.1	–	69.1	69.2
Ours (NeurIPS) [16]	36.1	37.7	37.8	70.2	82.0	82.6	35.3	37.7	37.9	58.8	66.1	66.4
Ours (Journal)	37.3	39.4	39.6	70.5	81.3	81.9	**35.6**	37.8	37.8	**61.1**	**68.0**	**68.2**

loss encourages the model to carve finer decision boundaries among semantically similar object classes, which is particularly beneficial in long-tailed regimes where confusing categories frequently dominate errors.

B. 3D Scene Graph Prediction with Object Co-occurrence

As assumption (2) and its empirical evidence in Sec. III-A, the distribution conditioned on object co-occurrence priors becomes flatter than the marginal distribution over all triplets. Despite this property, many prior approaches largely ignore co-occurrence structure and instead rely solely on model generalization. However, the role of co-occurrence is intuitive: if a scene contains objects such as `floor` or `table`, other objects in the same scene are more likely to exhibit relations such as `standing on`. Nevertheless, as shown in Appendix §.A, object co-occurrence itself is still imbalanced, and naively leveraging co-occurrence can therefore inherit long-tailed biases.

To address this issue, building upon our prior NeurIPS work, we introduce two complementary components: (i) Object-Centric Edge Updating (OEU) and (ii) Object Pair Inverse Frequency Weighting (OIW). The former explicitly conditions edge representations on the corresponding object pair to encourage learning of pair-dependent relation distributions $P(e_{ij} \mid o_i, o_j)$. The latter mitigates residual imbalance in object-pair statistics through reweighting. In this section, we describe the proposed object-centric edge updating scheme.

Object-Centric Edge Update. Our goal is to explicitly encourage the GNN to learn pair-conditional edge representations such that the relation distribution $P(e_{ij} \mid o_i, o_j)$ is modeled as a function of the object pair p_{ij}. To this end, we augment the standard GNN edge feature update with two additional update mechanisms that inject object pair information into the edge feature learning process.

Let $z_i^{n,t}$ and $z_j^{n,t}$ denote node features at the t-th GNN layer and let $z_{ij}^{e,t}$ denote the corresponding edge feature. First, we introduce a gated residual update that modulates how much the edge feature is updated based on the current object pair context:

$$\tilde{z}_{ij}^{e,t} = z_{ij}^{e,t-1} + w_{ij}^{n} \cdot (z_{ij}^{e,t} - z_{ij}^{e,t-1}) \tag{21}$$

where the gate w_{ij}^n is computed from the concatenated node features: $w_{ij}^n = \sigma\left(\text{MLP}(\text{CON}(z_i^{n,t}, z_j^{n,t}))\right)$. Here, $\text{CON}(\cdot,\cdot)$ denotes concatenation and $\sigma(\cdot)$ is the sigmoid activation. This update adaptively scales the effective edge update magnitude, enabling the model to apply stronger updates for object pairs that require more discriminative relation modeling.

Second, before feeding the edge feature into the subsequent GNN computation, we further condition the edge feature on the object pair using a feature-wise linear modulation mechanism inspired by FiLM [50]. Specifically, we compute modulation parameters from the node-pair features and apply them to the edge feature:

$$\gamma_{ij} = f_\theta(\text{CON}(z_i^{n,t}, z_j^{n,t})) \tag{22a}$$
$$\beta_{ij} = h_\phi(\text{CON}(z_i^{n,t}, z_j^{n,t})) \tag{22b}$$
$$z^{e,t+1} = \gamma_{ij} z^{e,t} + \beta_{ij} \tag{22c}$$

This modulation explicitly injects object-pair context into the edge feature representation, allowing the GNN to adjust edge semantics in a pair-dependent manner prior to message passing. We argue that these two update mechanisms jointly impose an explicit structural bias toward learning pair-conditional relation representations. Empirically, we demonstrate that incorporating these layers consistently improves scene graph prediction performance, validating the effectiveness of node-centric conditioning for exploiting object co-occurrence structure.

Object Pair-wise Re–balancing. As also evidenced by our empirical findings in Sec. III-A, the distribution over object pairs p_{ij} is itself imbalanced. To compensate for this bias in object co-occurrence statistics, we adopt an inverse frequency weighting scheme over object pairs, which we refer to as Object Pair Inverse Frequency Weighting (OIW). For each edge (i, j) associated with an object pair (o_i, o_j), we define the co-occurrence weight as

TABLE IV
QUANTITATIVE RESULTS (%) OF TOP-K MEAN RECALL (MR) ON 3DSSG DATASET WITH 160 ENTITY AND 27 RELATION CLASSES. FOR EACH SPLIT (HEAD, BODY, TAIL) WE REPORT *Predicate mR@3* AND *Predicate mR@5*. TRIPLET RECALL IS GIVEN FOR *Unseen* AND *Seen* RELATIONS.

Method	Predicate						Triplet			
	Head		Body		Tail		Unseen		Seen	
	mR@3	mR@5	mR@3	mR@5	mR@3	mR@5	R@50	R@100	R@50	R@100
SGPN [32]	**96.66**	99.17	66.19	85.73	10.18	28.41	15.78	29.60	66.60	77.03
SGFN [34]	95.08	**99.38**	70.02	87.81	38.67	58.21	22.59	35.68	71.44	80.11
VL-SAT [13]	96.31	99.21	80.03	93.64	52.38	66.13	31.28	47.26	75.09	82.25
Ours (NeurIPS) [16]	96.25	99.13	**84.92**	**95.34**	56.66	70.33	32.24	48.25	79.70	86.18
Ours (Journal)	95.63	99.27	84.01	93.89	**62.09**	**74.38**	**36.08**	**50.74**	**80.31**	**86.67**

$$w_{ij}^{co} = \left(\frac{(N(o_i) + C)(N(o_j) + C)}{N(o_i, o_j) + 1} \right)^{\alpha} \tag{23}$$

where $\alpha = 0.8$ and $N(o_i)$ denotes the frequency of object class o_i in the training set, and $N(o_i, o_j)$ denotes the frequency of the object pair (o_i, o_j) in the training set. C is the number of object classes, set to $C = 160$. Intuitively, this weighting re–balances highly frequent pairs while relatively emphasizing rare pairs, thereby reducing the dominance of head co-occurrences during predicate learning.

We incorporate the co-occurrence weight w_{ij}^{co} into the predicate classification objective, which is cast as a multi-label prediction problem. To ensure numerical stability and prevent excessively large gradients, we first normalize the weights to the range $[0, 1]$. We then re–balance the predicate loss by scaling the binary cross-entropy term of each edge with its corresponding co-occurrence weight:

$$\mathcal{L}_{\text{pred}} = \sum_{(i,j)\in\mathcal{R}} w_{ij}^{co} \text{BCE}\left(t_{ij}, y_{ij}\right) \tag{24}$$

where $\mathcal{R}$ denotes the set of edge indices within a mini-batch, t_{ij} is the multi-hot ground-truth predicate label, and y_{ij} is the predicted predicate probability (or logits, depending on the BCE implementation). The overall training objective follows our prior NeurIPS framework for the remaining components, with $\mathcal{L}_{\text{pred}}$ replacing the unweighted predicate loss.

V. EXPERIMENTS

TABLE V
QUANTITATIVE RESULTS (%) ON 3DSSG VALIDATION SET. THE **BOLD** DENOTES THE BEST PERFORMANCE ON 3DSSG DATASET WITH 160 ENTITY AND 27 RELATION CLASSES.

Model	Object		Predicate		Triplet	
	R@1	mR@1	R@1	mR@1	R@50	mR@50
SGPN [32]	49.46	10.86	86.92	32.01	85.38	41.52
SGFN [34]	53.36	21.24	89.00	46.68	88.59	70.82
VL-SAT [13]	55.93	21.41	89.81	54.03	89.35	65.09
Ours (NeurIPS) [16]	59.53	22.55	**91.27**	56.32	91.40	65.31
Ours (Journal)	**60.77**	**25.05**	90.26	**57.85**	**91.77**	**67.21**

A. Experimental Setup

We conduct experiments on the 3DSSG benchmark [32], a semantically annotated extension built upon 3RScan [48] for 3D semantic scene graph prediction [2]. The benchmark contains 1,553 real-world indoor scans and provides annotations for $|\mathcal{O}| = 160$ object categories and $|\mathcal{R}| = 26$ predicate classes across diverse household environments. We follow the official train/validation split released with the dataset and adhere to the standard preprocessing and evaluation conventions adopted in prior work. Additional implementation details are demonstrated in Appendix §.B.

B. Evaluation Metrics

Following the 3DSSG evaluation protocol [32], we report top-K *Recall* and *mean Recall* (R@K, mR@K) for object, predicate, and triplet recognition. Triplet confidence is computed by multiplying the predicted confidence scores of the subject, predicate, and object. A predicted triplet is considered correct only if all three labels match a ground-truth triplet. R@K is then computed by ranking triplets by confidence and measuring the fraction of ground-truth relations recovered within the top-K predictions.

In addition, we evaluate on the two standard tasks commonly used in scene graph literature [28]: *Scene Graph Classification* (SGCls), which predicts object categories and predicates given ground-truth object instances, and *Predicate Classification* (PredCls), which predicts predicates given ground-truth object categories and instances. For both tasks, we follow the widely adopted matching protocol [12], where a prediction is counted as correct if the predicted $\langle \text{subject}, \text{predicate}, \text{object} \rangle$ matches at least one annotated ground-truth relation. These metrics jointly capture object recognition quality and relational reasoning performance under standard 3DSSG settings.

C. Quantitative Results

Tab. III and IV report the quantitative comparisons on SGCls and PredCls with graph constraints, as well as performance

[2]We used the 3DSSG and 3RScan dataset [32] under the permission of its creators and authors. We contacted the authors by email and google form.

(a) VL-SAT (b) Ours, NeurIPS (c) Ours, Proposed

Fig. 6. t-SNE visualization of learned 3D object embeddings for 13 selected categories from the 3DSSG validation set. (a) VL-SAT exhibits substantial inter-class overlap and entangled embeddings. (b) Ours (NeurIPS) produces more structured clusters but still shows confusion among semantically similar classes. (c) Ours (Proposed) achieves the most discriminative embedding space with clearer cluster boundaries and reduced overlap, particularly for hard-negative pairs such as *table* vs. *couch table* and *cabinet* vs. *shelf*.

broken down by predicate frequency (Head/Body/Tail) and triplet split (Seen/Unseen). Although our method does not yet surpass the strongest prior approaches, it consistently improves upon our previous NeurIPS baseline [16], demonstrating that we successfully extend our earlier object-centric framework with the proposed analysis-driven components. We highlight with a gray shading on the metrics that most directly reflect and support our main claims in Tab. III and Tab. IV.

In particular, we observe consistent gains in PredCls Top-K mean Recall (mR@K) across evaluation in Table III. This improvement matches our analysis that the apparent predicate long-tail behavior is closely tied to object co-occurrence imbalance and that explicitly correcting pair-frequency bias benefits balanced predicate inference. The increase in PredCls mR@K therefore provides empirical evidence supporting the validity of our theoretical findings and the effectiveness of the proposed methodology.

In addition, Table V reports the component-wise classification performance on the 3DSSG validation set. As shown, our method improves not only predicate mean Recall, which is the primary target of our analysis, but also the overall recognition quality of scene graph components. In particular, we observe a notable increase in triplet mR@K, indicating that the gains in predicate prediction translate into more accurate relational triplets. These consistent improvements across predicates and triplets further support the validity of our theoretical analysis and corroborate that explicitly modeling object-centric priors can benefit long-tailed 3D scene graph prediction.

Moreover, Table IV highlights substantial improvements over the NeurIPS baseline on the most challenging regimes, notably Tail Predicates and Unseen Triplets. These gains indicate stronger generalization to rare relations and novel compositions, which is precisely where co-occurrence bias is most harmful. Collectively, these results support our central claim that long-tailed behavior in 3D scene graph prediction can be mitigated in an object-centric manner by explicitly modeling object co-occurrence and strengthening object representations, providing both empirical and theoretical justification for our approach. However, we observe a modest degradation on Head and parts of Body predicate performance, showing an inevitable head–tail trade-off when optimizing for balanced metrics. Nevertheless, the overall improvements on Tail Predicates and Unseen Triplets, together with the consistent gains in PredCls mR@K, indicate that this trade-off is acceptable in practice and aligned with our goal of improving long-tailed generalization.

D. *Qualitative Results*

Object Representation. To qualitatively evaluate the quality of learned 3D object representations, we visualize the embedding space using t-SNE over 13 selected object categories. We intentionally include semantically similar classes that frequently act as hard negatives in 3DSSG—such as `table` and `couch table`—where fine-grained discrimination is particularly challenging. As shown in Fig. 6.(a), Object encoder in VL-SAT [13] yields a largely entangled embedding space with substantial inter-class overlap, which is consistent with its weaker object classification. Our NeurIPS Object Feature Learning (OFL) baseline [16] in Fig. 6.(b) produces a more structured embedding space and improves separability; however, multiple categories remain partially overlapped, especially near the dense central region. In contrast, the proposed method in Fig. 6.(c) exhibits a more discriminative organization with clearer cluster boundaries and reduced overlap. For instance, `cabinet` and `shelf` form more distinct clusters than in the NeurIPS baseline. These qualitative observations support our goal of enhancing object discriminability by explicitly addressing hard negatives and instance-level variability, and provide intuitive evidence that the proposed approach strengthens object representations in the regimes most relevant to long-tailed 3D scene graph prediction.

Scene Graph Visualization. To qualitatively compare our proposed method with the NeurIPS baseline, we visualize predicted scene graphs alongside the ground truth on representative 3DSSG scenes (Fig. 7). Overall, the proposed model yields more faithful node and edge predictions, particularly in

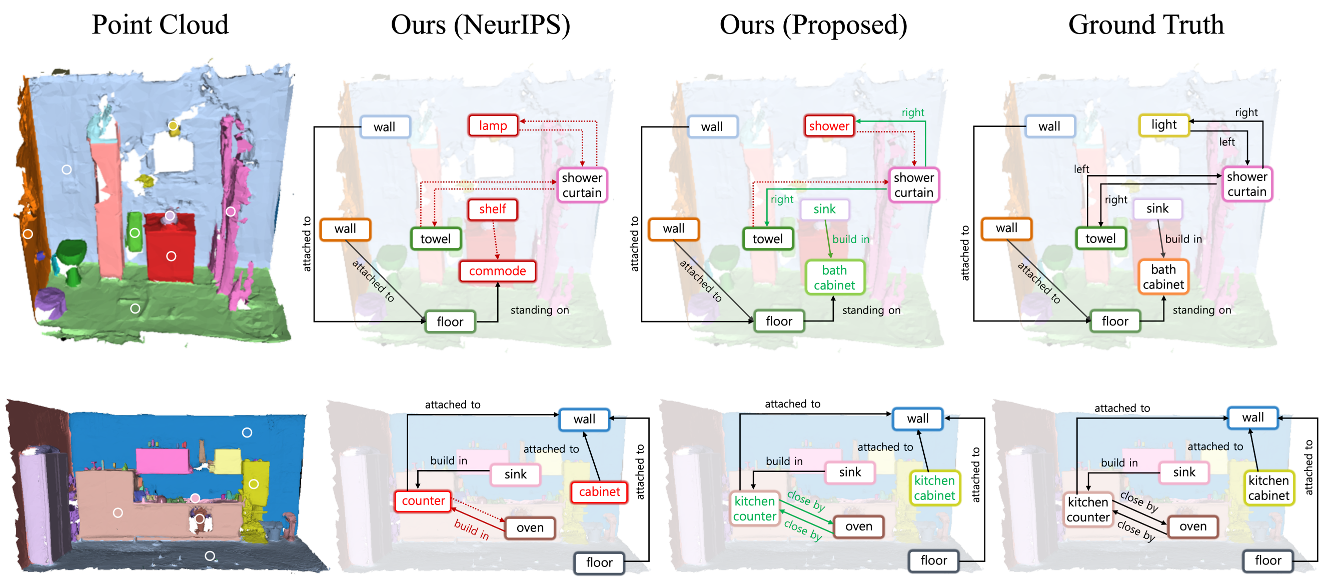

Fig. 7. Comparison of 3D object feature spaces via dimensionality reduction on 13 representative categories. Each point corresponds to a validation set instance. (a) VL-SAT produces overlapping clusters with weak inter-category separation. (b) Our NeurIPS method improves clustering structure but maintains residual confusion. (c) The proposed approach yields well-separated, compact clusters with minimal ambiguity between visually similar classes like *cabinet*/*shelf* and *table*/*couch table*.

TABLE VI
ABLATION STUDIES ON PROPOSED METHOD. ANALYZE IMPACT OF OBJECT CO-OCCURRENCE WEIGHT AND EDGE UPDATE GATING.

Model	SGCls			PredCls		
	mR@20	mR@50	mR@100	mR@20	mR@50	mR@100
Ours wo/ Both	33.5	35.6	35.6	59.9	66.6	66.9
Ours wo/ OEU	34.2	36.7	36.7	59.2	66.1	66.4
Ours wo/ OIW	34.4	37.2	37.3	60.8	**68.2**	**68.4**
Ours	**35.6**	**37.8**	**37.8**	**61.1**	68.0	68.2

TABLE VII
ABLATION STUDY ON OBJECT REPRESENTATION DISCRIMINABILITY (ORD) EVALUATED VIA LINEAR PROBING ON 3DSSG OBJECT FEATURES.

Model	Object Classification (Linear Probe)		
	Acc@1 ↑	Balanced Acc ↑	Macro-F1 ↑
Ours (NeurIPS)	54.70	22.14	33.84
Ours (NeurIPS)†	56.50	24.99	34.94
Ours w/o FGNR †	55.50	24.07	35.62
Ours †	**60.10**	**25.11**	**39.31**

† PointNet++ is used as the object feature encoder.

cases involving semantically similar object categories and rare relations.

First, we observe improved recognition of hard-negative object categories. In the bottom example, the NeurIPS baseline [16] misclassifies instances that belong to the confusing pair `kitchen counter` and `kitchen cabinet`, predicting them as the more generic labels `counter` and `cabinet`. In contrast, our method correctly distinguishes these fine-grained categories, which is consistent with the intended effect of Fine-Grained Negative Reweighting(FGNR) in the object encoder pretraining. Second, our method better captures informative, low-frequency predicates when they are coupled to rare object pairs. For instance, in the top example, the proposed model correctly predicts the tail predicate `build in` for the rare pair ⟨`sink`, `bath cabinet`⟩, whereas the baseline fails to recover this relation. These qualitative results support our empirical and theoretical findings that explicitly incorporating object-pair priors — via OEU and OIW — can improve long-tailed predicate inference beyond what stronger object features alone provide.

E. Ablation Study

Ablation on Object Co-occurrence. Table VI reports an ablation study on Object Pair Inverse Frequency Weighting (OIW) and Object-centric Edge Update (OEU). Removing both components yields a clear performance drop, confirming that the proposed modules provide non-trivial benefits.

OIW consistently improves SGCls mR, indicating that reweighting edges by object-pair co-occurrence effectively mitigates pair-frequency bias when the model is evaluated under full graph constraints. Although OIW alone leads to a marginal decrease in PredCls, this behavior suggests that directly re–balancing co-occurrence statistics may introduce a mild trade-off in predicate-only evaluation. Importantly, the gain in SGCls implies that OIW is particularly beneficial when object context and co-occurrence structure are jointly considered—precisely the regime our analysis targets—and it becomes more effective when combined with OEU. Overall, the full model achieves the best balance, supporting the view that OIW and OEU are complementary mechanisms for leveraging object-pair priors in 3D scene graph prediction.

Ablation on Object Representation Discriminability. Table VII reports an ablation study on Object Representation Discriminability (ORD) measured by linear probing on 3DSSG object features. We evaluate Top-1 accuracy, balanced accuracy, and macro-F1 to capture both overall recognition quality and class-balanced separability. Overall, each proposed component contributes positively, and the full model yields the strongest and most consistent gains across all metrics.

First, replacing the NeurIPS object encoder with PointNet++ improves ORD, increasing overall metrics by 1–2%. This indicates that a stronger geometric backbone alone provides a modest improvement in object separability. Next, removing Fine-Grained Negative Reweighting (w/o FGNR) degrades performance relative to the full model, suggesting that explicitly emphasizing confusing categories is important for sharpening the representation space – while w/o FGNR improves over the NeurIPS baselines, it underperforms the full model in all metrics. Finally, our full approach achieves the best results with huge margin, demonstrating that the proposed pretraining strategy substantially enhances object-level discriminability beyond what can be obtained by simply changing the encoder architecture. These results provide empirical support that our design choices effectively increase ORD, particularly in a class-balanced sense, which is critical for long-tailed 3D scene graph prediction.

VI. Conclusion

In this paper, we revisited long-tailed 3D semantic scene graph prediction from an object-centric perspective and showed that a purely predicate-centered view can be misleading under closed-vocabulary imbalanced benchmarks. Extending our prior framework [16], we first provide a key empirical and theoretical findings – improving the 3D Object Representation Discriminability(ORD) alone does not sufficiently resolve predicate long-tail distribution problems.

We further identify object co-occurrence imbalance as a primary factor underlying predicate imbalance. Our theoretical analysis explains how, once object posteriors become highly confident, optimization becomes increasingly dominated by head object pairs, inducing a pair-frequency gradient bias that conflicts with balanced objectives. Guided by these insights, we propose ORD enhancement methods, Object-centric Edge Updating, and object pair-wise re-balancing. Experiments demonstrate consistent gains in PredCls mR@K, with notable improvements on tail predicates and unseen triplets, supporting the validity of our analysis. Overall, our results suggest that long-tailed behavior in 3D scene graph prediction can be effectively mitigated through an object-centric scheme of co-occurrence priors. We believe our work motivates future research on 3D semantic scene graph prediction, particularly toward improving the estimation of scarce predicates.

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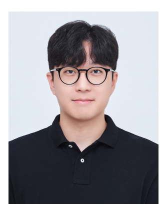

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